

MAPLE LAKE, MN, USA

WMO: 722144

Lat: 45.236N

Lon: 93.986W

Elev: 1028

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-8.5	-19.3	2.0	-12.0	-15.3	2.5	-7.9	22.6	13.9	19.8	17.8	4.6	300	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.2	73.8	85.7	72.5	82.7	70.3	77.0	84.6	75.0	82.5	73.0	80.1	7.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	135.2	82.4	72.5	124.9	80.1	70.6	117.1	77.8	41.2	84.7	39.2	82.8	37.3	80.1	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.8	17.6	15.7	-21.0	93.8	5.3	2.9	-24.7	95.8	-27.8	97.5	-30.8	99.2	-34.6	101.3		
			-21.3	80.3	5.2	2.3	-25.0	82.0	-28.0	83.4	-30.9	84.7	-34.7	86.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.8	14.6	16.9	31.5	45.0	57.6	67.5	71.8	68.6	61.8	47.5	33.8	19.4
	DBStd	22.40	13.19	11.92	12.37	9.91	8.90	6.18	5.37	5.54	8.17	8.80	10.40	11.70
	HDD50	4446	1096	928	584	207	28	0	0	0	7	154	492	950
	HDD65	7945	1561	1348	1039	601	264	43	9	26	157	546	936	1415
	CDD50	2552	0	0	11	57	263	525	675	578	360	77	6	0
	CDD65	575	0	0	0	1	34	118	219	139	60	4	0	0
	CDH74	4830	0	0	2	23	385	962	1904	1064	448	42	0	0
	CDH80	1289	0	0	0	1	119	256	561	247	100	5	0	0
	WSAvg	6.2	6.7	7.0	6.8	7.9	6.8	5.5	4.5	4.2	5.2	6.5	6.8	6.5
Precipitation	PrecAvg	31.4	0.7	0.7	1.5	2.8	3.7	4.8	3.9	4.3	3.6	2.6	1.6	0.9
	PrecMax	50.9	2.0	1.7	3.6	8.0	7.7	13.3	10.4	8.5	8.3	6.6	4.7	2.2
	PrecMin	21.5	0.1	0.1	0.3	0.6	0.8	1.9	1.0	0.7	0.6	0.6	0.1	0.2
	PrecStd	6.2	0.6	0.4	0.8	1.7	1.7	2.5	2.2	2.1	2.2	1.6	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	51.7	68.8	77.4	90.4	90.7	92.3	89.0	87.4	79.4	64.9	46.9
		MCWB	37.7	48.1	57.3	59.0	68.4	73.3	78.0	75.6	73.7	63.6	54.7	43.5
	2%	DB	38.1	41.7	60.4	70.6	82.6	86.2	88.3	85.7	82.9	72.6	58.7	40.6
		MCWB	35.2	37.5	52.6	55.0	65.0	71.3	74.9	74.2	70.6	60.3	50.2	37.2
	5%	DB	35.2	37.1	54.1	66.3	78.1	82.5	85.8	82.4	79.6	67.0	52.8	36.9
		MCWB	33.7	34.3	48.8	53.5	63.1	68.9	73.4	70.8	68.8	57.5	46.7	35.1
	10%	DB	32.3	33.4	47.9	61.8	73.0	79.4	82.8	80.2	75.5	61.8	48.2	34.3
		MCWB	31.2	31.4	43.1	50.4	60.7	67.0	71.5	69.1	66.9	54.1	43.7	33.2
	0.4%	WB	38.1	47.9	61.2	61.0	73.0	77.4	80.8	78.3	76.9	68.5	57.1	44.7
		MCDB	41.9	52.2	66.9	72.9	85.3	85.9	87.4	86.2	83.7	73.2	63.7	46.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	38.3	54.1	57.8	68.6	74.2	77.7	76.1	73.3	62.3	51.6	37.6
		MCDB	37.7	41.4	59.2	66.8	76.9	82.3	85.3	83.1	79.7	69.7	57.7	39.7
	5%	WB	33.5	34.6	48.3	54.8	65.7	71.3	75.5	73.8	70.7	58.8	47.2	35.2
		MCDB	34.9	36.9	54.4	63.9	74.3	79.7	83.5	80.1	76.4	65.4	52.0	36.6
	10%	WB	31.3	31.8	43.2	51.7	62.7	69.1	73.4	71.6	68.2	55.1	43.8	33.1
		MCDB	32.3	33.4	47.5	60.1	71.0	76.6	81.0	77.1	74.7	60.7	48.3	34.2
Mean Daily Temperature Range	5% DB	MDBR	15.5	17.7	17.1	19.0	20.2	18.7	19.4	19.0	19.5	17.2	15.0	13.6
		MCDBR	16.9	19.3	23.7	27.4	27.1	22.6	21.3	21.1	22.6	25.8	22.1	14.7
		MCWBR	14.3	15.6	15.5	15.4	13.2	11.1	10.6	10.8	12.0	14.9	14.5	12.4
	5% WB	MCDBR	16.6	18.8	21.9	23.7	22.8	19.2	19.0	18.4	19.9	21.4	20.3	13.7
		MCWBR	14.3	15.4	15.1	14.8	12.5	11.0	10.9	10.9	12.3	14.1	14.4	11.8
Clear-Sky Solar Irradiance	taub		0.264	0.276	0.311	0.367	0.401	0.392	0.389	0.400	0.368	0.315	0.286	0.261
	taud		2.415	2.373	2.395	2.311	2.271	2.342	2.370	2.334	2.392	2.536	2.514	2.444
	Ebn at Noon		273	290	293	283	275	278	277	270	270	273	264	263
	Edh at Noon		23	28	32	38	42	39	37	37	32	24	21	20
All-Sky Solar Radiation	RadAvg		549	893	1208	1474	1701	1891	2016	1678	1299	823	525	386
	RadStd		66	94	110	137	136	139	71	129	94	113	49	47

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page